

## EXPERIMENT NO. 3

### WEBSITE MIRRORING USING HTTRACK

#### Aim

To familiarize students with the working of the HTTrack website mirroring tool.

#### Introduction / Description

HTTrack is an easy-to-use offline browser utility that allows a website to be downloaded for offline viewing. It can download web pages and associated resources while preserving the website structure. The downloaded copy can be examined locally without repeatedly accessing the original website.

#### Uses of HTTrack

- Offline browsing of downloaded website content.
- Website structure and content analysis.
- Educational and practical study of website resources.
- Supporting authorised digital investigation and forensic analysis.

#### Important Features

- Downloads accessible web pages and associated resources.
- Preserves relative links and directory structure where possible.
- Provides options for controlling the mirror operation.
- Can continue or interrupt a download according to the selected options.
- Creates a local project containing the mirrored website.

#### Requirements

**Windows:** Windows computer, HTTrack and an Internet/network connection.

**Kali Linux:** Kali Linux, HTTrack package and Terminal.

#### PROCEDURE

##### Step 1 – Create a Project

Open HTTrack and create a new project. Enter an appropriate project name and select the local directory in which the website will be stored. An existing project can also be selected when continuing an earlier operation. Click NEXT to proceed.

##### Step 2 – Enter the Website Address

Select the required action and enter the URL of the authorised website to be mirrored. Configure the available options according to the required scope of the download and proceed to the next stage.

##### Step 3 – Ready to Start

Connect the system to the Internet. Check the selected URL and project directory. Select the appropriate download action and continue. HTTrack prepares the project and begins the mirror process.

##### Step 4 – Start the Mirror

Start the operation and allow HTTrack to download the accessible resources. The progress window provides information about the current operation. The process can be stopped when required.

### **Step 5 – Finish**

After the required pages and resources have been downloaded, select the finish option. HTTrack stores the resulting website in the selected project directory.

### **Step 6 – Check the Result**

Open the destination directory and inspect the downloaded files. The local website can be opened using a browser to verify the pages and links that were successfully mirrored.

## Linux Installation

On a Debian-based system, HTTrack can be installed using the package manager when the package is available:

```
sudo apt update
sudo apt install httrack
httrack --version
```

## Download / Mirror Options

The record discusses options such as mirroring the required website, following links, continuing interrupted downloads and checking the downloaded structure. These settings should be selected according to the required practical task.

## Downloaded Content

After completion, the project directory can contain HTML pages, images, stylesheets and other accessible resources. HTTrack creates a local representation of the website and attempts to maintain the relative structure required for browsing.

## Observation

- The website was successfully selected for mirroring.
- The project and destination directory were created.
- Accessible pages and resources were downloaded.
- The downloaded files were stored in the selected local directory.
- Relative links were maintained where possible.
- The mirrored site could be examined locally.
- Website pages, images, CSS and other accessible resources may be present in the downloaded copy.

## Analysis

The experiment produces an offline copy of the selected website. The mirrored structure contains downloaded HTML pages and supporting resources. The local copy can then be inspected without repeatedly accessing the original website.

HTTrack creates a local representation of the website and attempts to preserve the relative structure required for browsing. The resulting project directory can therefore be examined to understand the downloaded pages, resources and directory organisation.

## Limitations

- Dynamic content may not behave exactly like the original website.
- Restricted or protected resources may not be downloaded.
- Content requiring server-side processing may not work in the local copy.
- Changes made to the original website after mirroring are not automatically reflected.

## Forensic Relevance

Website mirroring can assist authorised investigation by preserving accessible website material in a local project. The local copy can be inspected for pages, resources and structure without repeatedly accessing

the original website.

## Precautions

- 1 Only websites and networks for which the user has permission should be mirrored.
- 2 The download scope should be controlled so that unnecessary content is not collected.
- 3 Use an authorised target for the practical experiment.
- 4 Check the selected URL and destination directory before starting the mirror.
- 5 Review the downloaded structure after the operation is completed.

## Result Analysis

The selected authorised website was successfully mirrored using HTTrack. The tool created a local project containing the accessible website pages, resources and directory structure. The downloaded content could be examined locally, allowing the structure of the website to be studied without repeatedly accessing the original website.

The result depends on the structure of the website, access restrictions and the options selected during the mirroring process. Accessible resources such as HTML pages, images, CSS and other files may be included, while dynamic, protected or server-side content may not behave in the same way in the local copy.

## Result

Successfully mirrored the authorised website using HTTrack and created an offline copy of its accessible content and directory structure.

## Conclusion

HTTrack provides a practical method for creating an offline website copy. It is useful for learning website structure, offline browsing and authorised forensic analysis. Its project-based downloading and configurable mirroring options allow accessible website material to be stored and examined locally.

## Practical Flow

**Open HTTrack → Create Project → Enter URL → Select Mirror/Download Options → Start Download → Wait for Completion → Open Destination Directory → Verify the Mirrored Website**

## Final Observations

- 1 HTTrack can create offline copies of authorised websites.
- 2 HTML pages can be downloaded into the local project.
- 3 Images and other accessible resources can also be downloaded.
- 4 The downloaded files are stored in the selected destination directory.
- 5 Relative links and directory structure are maintained where possible.
- 6 The mirrored website can be examined locally.
- 7 Dynamic or protected content may not be completely mirrored.
- 8 The scope of mirroring should be controlled during an authorised practical exercise.